Impact energy and TNT equivalents

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Convert an energy estimate without claiming a complete damage prediction.

SUCCESS CRITERIA - I can...

1. Convert speed units before squaring.
2. Calculate kinetic energy from the supplied mass and velocity.
3. Distinguish an energy equivalent from an explosion or damage forecast.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A constant **30 N** forward force acts at steady **5 m/s** for **7 s**. **Find mechanical power** and work.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $K = \frac{1}{2}mv^2$
TNT equivalent (tons) = $K/(\mathbf{4.184\times10^9\ J})$.
1 ton TNT equivalent = $\mathbf{4.184\times10^9\ J}$. Equal energy does not imply identical effects.

Worked reasoning: A fictional event transfers $\mathbf{8.4\times10^6\ J}$. Using $\mathbf{4.2\times10^6\ J}$ per kg TNT as an energy comparison gives **2 kg** TNT equivalent. This compares energy, not identical effects.

Your turn: Draw a labeled energy comparison and **name one effect this number** alone cannot predict.

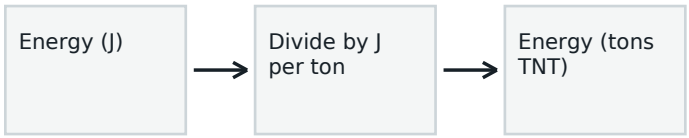
Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two reports describe the same fictional impact: one uses joules and the other uses TNT-equivalent tons.
1. Use the conversion diagram to **explain why the units can** express the same quantity.
2. What does this comparison leave out?



Marzano check - circle: ① ② ③

04 Apply the physics

A hypothetical $\mathbf{10^6\ kg}$ asteroid moves at **10 km/s**. Convert speed to m/s, calculate kinetic energy and express that energy in TNT-equivalent tons using $\mathbf{4.184\times10^9\ J}$ per ton. **Complete the conversion table**.

Energy (J)	J per ton TNT	Equivalent tons
___	4.184×10^9	___

Marzano check - circle: ① ② ③

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Continue down each column.

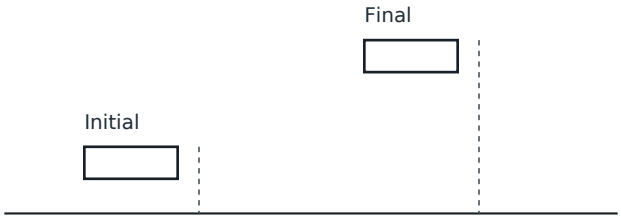
05 Explain your reasoning

Why should an energy-equivalent number not be reported as a precise area of destruction?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **5 kg** object descends **7 m** near Earth. Use g=**10**. Find ΔU and the work done by gravity.
Label the distances or energy levels used in your reasoning.



Choose and label a zero-height level.

Marzano check - circle: 1 2 3

07 Exit check

At fixed speed, mass doubles while assumed energy coupling fraction halves.
1. **Compare coupled energy**.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **2000 kg** hypothetical object moves at **1000 m/s**.
1. **Find kinetic energy** and kg TNT equivalent using **4.2×10^{6} J/kg**.
2. **Explain why neither alone predicts** a crater size.
3. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Total energy → Area / coupling → Local effects

Marzano check - circle: 1 2 3